

Wave Release Coordination under Vertical Resource Constraints in Multi-Story AMR Warehouse

Abstract

In multi-story fulfillment warehouses, autonomous mobile robots (AMRs) are confined to floors and share a limited number of freight elevators that connect the levels. Because elevator capacity is the system’s binding constraint, the upstream decision of which orders to release together (the wave composition) determines how difficult each downstream AMR elevator dispatch problem is. The joint study of wave composition and elevator capacity coupling in the literature on multi-story AMR warehouses remains an open research direction. We formalize this gap as the wave release coordination problem under vertical resource constraints, a two-stage scheduler that decouples tactical wave composition from operational AMR-elevator dispatch. On this scaffold, we develop two analytical tools. The first is a Bound-and-Gap decomposition theorem that splits the value of structural information into two non-negative components on any finite partition of a feature representation, instantiated here on a three-dimensional $\Phi = (C, I, T)$ of wave composition, namely vertical activity diversity, directional imbalance, and temporal clustering; the theorem splits the value of wave-structure information into two non-negative components. The second is a Model-Dominance Hedge Rule that selects a single dispatch decision robust to elevator-model uncertainty under a chain of per-wave stochastic dominance conditions across the family of elevator models $\{M_1, M_2, M_3\}$, accompanied by a closed-form worst-case loss expression against any model-specific optimum. Both tools are validated by a pre-registered, publication-scale simulation study. The framework then delivers two managerial outcomes: from the decomposition, (i) a structural account of the value of wave design, separating a partition-intrinsic upper-tail component from a component a better policy can recover; and from the hedge rule, (ii) a closed-form dispatch policy that operators apply without identifying the true elevator model online, together with a closed-form expression for its worst-case loss against any model-specific optimum.

Introduction

The growth of e-commerce and the demand for next-day and same-day delivery have led to the relocation of distribution centers nearer to dense urban areas, where land availability is limited, and expansion is restricted by zoning laws and high costs. Operators increasingly resort to multi-story fulfillment buildings, stacking warehouse floors vertically because a flat facility with equivalent throughput would have to be located far from urban customers. These buildings usually have autonomous mobile robots (AMRs) managing horizontal movement on each floor. A few freight elevators connect levels and manage all floor-to-floor transfers for the fleet. The main bottleneck has shifted from floor space or the number of robots to elevator capacity, causing dozens of AMRs to sometimes remain idle while waiting for limited elevator trips.

Operators manage this congestion by releasing orders in waves, with short release windows during which a chosen subset of orders is dispatched together. The composition of each wave (which orders to bundle) directly determines how many vertical transitions the elevators must serve, in which directions, and with what temporal stagger; it therefore sets the difficulty of the downstream dispatch

problem before any robot moves. In current warehouse practice, wave composition is set heuristically by destination floor, carrier cutoff, or arrival order, because no framework links this tactical decision to its operational consequences in the context of shared-elevator coupling. Yet the wave layer is the most cost-effective way to relieve a vertical bottleneck: re-batching already-released orders does not change either the floor layout or the fleet size. Despite this leverage, these four elements (wave composition, multi-story deployment, flexible AMR fleets, and shared-elevator capacity) have so far been studied separately; their joint coupling defines a problem class that has not yet been formalized as a single decision problem.

Prior work touches on the coupling but never frames it as a single decision problem. The closest precedent, multi-agent robot-lift scheduling (Chakravarty et al., 2025), treats the lift as a constraint to be satisfied under fixed wave content, leaving wave composition exogenous; the same is true at the device level for vertical lift module batching (Nicolas et al., 2018). Multi-story Robotic Mobile Fulfillment System (RMFS) work avoids the coupling altogether by confining robots to a tier through captive shuttles (Wu et al., 2024) or by treating levels as isolated zones (Tadumadze et al., 2023). Planar AMR fulfillment optimizes order grouping and wave cardinality (Qin et al., 2024; Scholz et al., 2017a; Žulj et al., 2022a) but never against a shared vertical resource. The wave–elevator coupling thus falls between three established threads rather than within any of them. Two methodological lines inform our framing, but do not close this gap. Prediction-to-decision regret (Elmachtoub & Grigas, 2022; Vera et al., 2020) and robust scheduling under model uncertainty (Lu & Shen, 2021; Wiesemann et al., 2013a) provide tools we draw on, but neither targets the wave–elevator coupling. Experiments across typical operating regimes surface two structural features that our methodology must address. First, wave composition has a small but systematic effect on make-span, and the direction of that effect varies across regimes, so any framework must measure regime-conditional value rather than report a single global gain. Second, under the natural family of elevator models $\mathcal{M}$, a per-wave stochastic dominance order between the candidate models holds with high empirical frequency, opening the possibility of a closed-form decision rule that does not require online model identification. Specific magnitudes and dominance frequencies are reported in Section 5.

We develop two analytical tools to address these features. The first is a Bound-and-Gap framework: a decomposition theorem that splits the value of wave-structure information into two non-negative components, namely a partition-intrinsic upper-tail component and a recoverable policy component. The second is a Model-Dominance Hedge Rule: a closed-form dispatch policy that handles uncertainty over which elevator model best describes a warehouse, telling operators which wave to release without identifying the true model online and bounding the cost of being wrong.

The contributions of this paper are threefold.

Problem formulation. We formalize the wave-release coordination problem under vertical resource constraints as a two-stage scheduler that decouples tactical wave composition from operational AMR–elevator execution, subject to a hard per-trip elevator-capacity bound. Existing work treats wave composition, multi-story deployment, flexible AMR fleets, and shared-elevator capacity as separate concerns; this formulation integrates them into a single coupled system, revealing the wave–elevator interaction that no individual strand captures.

Methodology. We develop two analytical tools and prove each as a theorem. The first, a Bound-and-Gap decomposition theorem, splits the value of wave-structure information into a partition-intrinsic

upper-tail component and a recoverable policy component on a structured representation $\Phi = (C, I, T)$ of wave composition; we show by equivalence that the policy component coincides with the Smart-Predict-then-Optimize regret of the partition-constant predictor class, embedding our decomposition in the prediction-to-decision regret literature. The second, a Model-Dominance Hedge Rule, establishes a minimax-collapse result under per-wave stochastic dominance and reduces the dispatch decision over competing elevator models to a closed-form policy; we show by equivalence that, under chain dominance, this policy coincides with the Wasserstein distributionally robust solution at an appropriately sized ambiguity radius. We position the framework as diagnostic rather than prescriptive: it characterizes where structural information adds value and bounds the cost of model misspecification, but it is not proposed as a competitive solver for wave composition; a local-search optimizer outperforms its policy realization, and we report this gap honestly in Section 5.

Managerial implications. Both tools are validated by pre-registered simulation. From the Bound-and-Gap decomposition, we obtain a structural account of the value of wave-structure information across operating regimes, separating the partition-intrinsic headroom from the component that a better policy can recover. From the Hedge Rule, we obtain a closed-form dispatch policy that operators can apply without identifying the true elevator model online, with a closed-form worst-case loss expression relative to any model-specific optimum; the policy's domain of validity extends beyond the low-capacity regime in which the rule was derived.

Together, these contributions reframe wave release in multi-story AMR warehouses as a structured engineering decision rather than a heuristic one. The operational implication is that the same warehouse data that informs heuristic batching decisions can, with these tools, support a quantified account of where structured wave design adds value and of the worst-case cost of model misspecification.

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 formulates the two-stage wave release coordination problem as a simulation-based decision problem with a tractable analytical layer. Section 4 develops the two tools that operate on this layer. Section 5 presents computational experiments. Section 6 discusses managerial implications and concludes.

Related works

The wave–elevator coupling we study sits between several established bodies of work, each touching on a piece of our setting without addressing the joint problem. We position our contribution against the literature in three steps: first we summarize the broader warehouse-OR landscape through three recent surveys; then we walk through the four problem-class threads whose intersection defines our gap—multi-story AMR and elevator group control, wave and order batching as a tactical decision, tier-captive multi-story RMFS, and data-driven fleet coordination—reading each thread for what it supplies and what it omits relative to our setting. Finally, we summarize three methodological strands on which our two analytical tools rest, with precise equivalences established in Section 4.

Three surveys frame the landscape. Boysen et al. (2019) survey warehousing in the e-commerce era, identifying the storage-and-retrieval challenges that motivate vertically stacked fulfilment buildings; Azadeh, De Koster, et al. (2019) review robotized, and automated warehouse systems with a dedicated treatment of robotic mobile fulfilment; and Pardo et al. (2024) provide the most recent

taxonomy of the order-batching family of problems, classifying decisions by their strategic, tactical, or operational layer. Within the scope of these surveys, we have not found the multi-story wave-elevator coupling treated as a standalone problem class, consistent with our positioning of this work as a formalization step.

The physical infrastructure of our setting, floors connected by shared freight elevators, has been studied piecewise but never jointly. Multi-agent robot-lift scheduling on a fixed task set is the closest precedent at the elevator layer: Chakravarty et al. (2025) compute provably optimal lift schedules for AMR fleets via Boolean satisfiability, achieving exact optimality on hotel and hospital deployments. Their formulation takes the robot-task assignment as exogenous and finds the time-optimal schedule on a fixed task set; in our setting, that fix is what we must decide, because the composition of the task set determines downstream elevator concurrency and is the dominant tactical lever for make-span. We treat wave composition as an upstream decision variable that generates the task set their methods would then optimally schedule, with the two layers coupled through shared elevator capacity. The elevator subsystem itself has a parallel literature in elevator group control: Crites et al. (1998a) remain the canonical reference for learned dispatch policies under stochastic passenger arrivals, and Tsai et al. (2025) extend the family to standby strategies in smart buildings. These works optimize the elevator dispatch given an exogenous arrival process; we optimize that process by choosing the wave that generates it. At the storage layer, Azadeh, Roy, et al. (2019) analyze vertical robotic storage and retrieval systems, but the vertical degree of freedom there is robot-internal rather than building-shared. Multi-story robotic mobile fulfilment provides a complementary thread: Wu et al. (2024) address inbound job scheduling in four-way shuttle systems, and Tadumadze et al. (2023) study order and pod assignment in multi-level RMFS. Both adopt a tier-captive transport model in which each shuttle or pod is assigned to a specific layer, so layer-level operations decouple, and wave composition does not appear as a decision variable. In our setting, by contrast, flexible AMRs queue at a shared elevator pool, so orders released to one floor consume vertical capacity that orders on other floors are competing for; multi-layer operations cannot be solved layer by layer.

Treating the composition of an order batch, rather than its cardinality alone, as a decision variable has a long arc in warehouse OR, but the entire arc has been developed at a single-floor scale. Gademann et al. (2001) introduced wave picking in parallel-aisle warehouses with a make-span objective, the closest precedent to ours in objective function; Bozer & Kile (2008) and the canonical treatment of Bartholdi & Hackman (2019) embed the decision in walk-and-pick systems. Ardjmand et al. (2018) minimize order-picking make-span with multiple pickers under wave picking, an objective identical to ours, but on a single floor without a shared vertical resource. Rasmi et al. (2022) extend wave picking to the mixed-shelves storage strategy in e-commerce warehouses; Schiffer et al. (2022) derive optimal picking policies in the same e-commerce setting; Haouassi et al. (2022) integrate order line batching with batch scheduling and picker routing under split-order flexibility. Modern AMR-assisted variants enlarge the decision class: Scholz et al. (2017b) formalize the Joint Order Batching, Assignment, Sequencing, and Picker Routing Problem; Žulj et al. (2022b) lift it to AMR-assisted picker-to-parts systems; Qin et al. (2024) study multi-tote AMR fulfilment and identify an empirical wave-size optimum near 100 orders, where wave size is a cardinality parameter rather than a composition decision. Two device-level batching threads are adjacent rather than parent: Nicolas et al. (2018) formulate order batching for vertical lift modules where each module is a single mechanical

device and the decision is which order lines to retrieve in one tray load, and Boysen et al. (2018) address the minimum order spread sequencing problem in automated sorting; the mathematics in both is internal to the device and does not generalize to a multi-elevator capacity shared across a horizontal AMR fleet. The Pardo et al. (2024) taxonomy classifies all of these works at the tactical layer of order-batching decisions, and our wave-composition decision belongs to the same layer; what is new in our setting is that the tactical decision is evaluated against a shared vertical resource that serves a building-wide AMR fleet. We extend this decision class to the multi-story setting and represent wave composition through $\Phi = (C, I, T)$, where each axis captures a distinct vertical-resource stress.

A parallel research stream addresses warehouse fleet control through learning rather than formal optimization, and we summarize it here to position our contribution as complementary rather than competitive. Crites et al. (1998b), already mentioned for elevator group control, established the foundational reinforcement-learning treatment of vertical-resource scheduling; the line continues today in AMR-specific work. Ma et al. (2025) propose an end-to-end deep reinforcement-learning framework for modular task allocation in autonomous mobile systems and report state-of-the-art performance on multi-robot dispatching benchmarks; Dhanaraj et al. (2025) study proactive contingency-aware task allocation and scheduling in multi-robot multi-human cells via hindsight optimization; Wen & Ma (2024) develop an indicator-based evolutionary algorithm for multi-objective multi-robot task allocation; Wesselhöft et al. (2022) survey reinforcement-learning approaches to AMR fleet control and identify six open challenges, including the sim-to-real gap and data efficiency. These approaches obtain policies without an explicit elevator-model commitment and would, in principle, sidestep the model uncertainty problem that our Hedge Rule addresses. We see two reasons the analytical route is complementary rather than dominated. First, the diagnostic value of our Bound-and-Gap decomposition does not transfer to learned policies, which optimize a single scalar objective and do not expose the partition-intrinsic-versus-recoverable split. Second, the closed-form Hedge Rule requires no training data and provides a deployment-ready reference against which any learned policy can be measured. The two paradigms, therefore, answer different questions about the same physical system.

Three threads of prior methodological work inform our analytical apparatus: the first two are connected to our tools by precise equivalences established in Section 4, and the third by a structural contrast. First, the prediction-to-decision regret literature (Chenreddy & Delage, 2024; Elmachoub & Grigas, 2020; Vera et al., 2020) bounds the algorithmic loss when a learned predictor drives a downstream optimizer. Theorem 1 in Section 4.1.2 gives this regret a partition perspective: the policy-slack component M_Φ of our decomposition equals the Smart-Predict-then-Optimize regret of the partition-constant predictor class, so the literature’s training-time regret object and our post-hoc, structured-feature decomposition refer to the same underlying quantity on a finite partition. Our contribution is the decomposition itself, which separates this SPO regret from an independent partition-intrinsic headroom term H_{up} . Second, Wasserstein distributionally robust optimization (Blanchet & Murthy, 2019; Gao & Kleywegt, 2023; Mohajerin Esfahani & Kuhn, 2018) hedges decisions against worst-case distributions within a transport-distance ambiguity ball of a nominal model, typically reformulated as a convex program; the underlying robust-optimization framework dates to Delage & Ye (2010) and is surveyed by Bertsimas et al. (2011). Theorem 2 in Section 4.2.2 shows that under a chain of per-wave stochastic dominance conditions, our minimax wave-corner

selection over $\{M_1, M_2, M_3\}$ coincides with the Wasserstein distributionally robust solution at the appropriate ambiguity radius, collapsing the convex program into a closed-form one-line rule. Our contribution is the dominance-conditional collapse, which exploits an empirically verified structural property to replace optimisation with identification. Third, robust scheduling under model uncertainty (Lu & Shen, 2021; Wiesemann et al., 2013b) hedges against parametric variation within a single agreed-upon model class. Our elevator-modelling uncertainty is structurally different: throughput aggregation and true co-occupancy batching are qualitatively distinct models, not parameterizations of one family. The Model-Dominance Hedge Rule addresses this structural model-class disagreement rather than parametric uncertainty within a class.

Taken together, the four problem-class threads each address one aspect of the wave–elevator coupling, but none integrates the full four-way interaction of wave composition, multi-story deployment, flexible AMR fleets, and shared-elevator capacity. The methodological threads provide analytical grounding for our two tools: the first two supply the frameworks our equivalence theorems extend, and the third positions them against a competing approach. The empirical findings demonstrating these contributions in action are deferred to Sections 5 and 6. We formalize the joint problem class in Section 3 and develop the two analytical tools for it in Section 4.

Problem Formulation

We consider a multi-story fulfillment warehouse with F floors connected by E shared freight elevators. A fleet of $|A|$ floor-bound autonomous mobile robots (AMRs) handle horizontal travel on each floor; floor-to-floor transfers go through the elevators, with each elevator carrying at most c AMRs per trip. Orders arrive over a planning horizon, each with a known source floor, destination floor, and release time. A wave is a subset of orders released into the system simultaneously at the start of a release window of length Δ . The system performance metric is the wave make-span: the time at which the last order in the wave is delivered, given a fixed operational dispatch policy.

This setting decomposes naturally into two coupled decision layers. The tactical layer chooses which orders to bundle into a wave W , namely the wave composition decision; this is the focus of our analysis. The operational layer determines, given W , which AMR serves which order, which AMRs board each elevator trip, and in what order, decisions we delegate to a deterministic dispatch policy, defaulting to capacity-bounded FIFO boarding under elevator model $M \in \mathcal{M}$, with full simulator details given in Section 4. Section 5 additionally evaluates a destination-clustered alternative as an experimental treatment factor, not as an additional decision variable of the formulation. The two layers are coupled through the make-span: a wave whose composition is structured to ease the downstream dispatch yields a smaller make-span than an unstructured wave of the same size. Section 3 first lists the sets, parameters, decision variable, and assumptions on which the formulation rests, then defines a structured-feature representation Φ of wave composition and the family of elevator models $\mathcal{M}$, and concludes with the objective and constraints.

Sets and indexes

$\mathcal{O}$: set of all orders over the planning horizon, indexed by o .

$\mathcal{F} = \{1, 2, \dots, F\}$: set of floors, indexed by f .

A : AMR fleet, indexed by a .

$\mathcal{E}$: set of elevators, indexed by e .

W : sequence of waves over the planning horizon, indexed by ω .

$t \in \mathbb{R}_{\geq 0}$: continuous time.

$\mathcal{M} = \{M_1, M_2, M_3\}$: elevator-model set, indexed by m .

Parameters

We group parameters by category to make their roles explicit.

Physical infrastructure (deterministic):

$F \in \mathbb{Z}_{\geq 2}$: number of floors

$|A|$: AMR fleet size

E : number of elevators

$c \in \mathbb{Z}_{\geq 1}$: per-trip elevator capacity (number of AMRs per elevator trip)

Timing primitives (deterministic under M_1, M_2 ; stochastic under M_3):

$\tau_s \in \mathbb{R}_{>0}$: per-event AMR service time at pickup or dropoff

$\tau_e \in \mathbb{R}_{>0}$: per-floor elevator travel time

$\tau_d \in \mathbb{R}_{>0}$: elevator dwell time per stop

Order attributes (input data):

$s_o \in \mathcal{F}$: source floor of order o

$d_o \in \mathcal{F}$: destination floor of order o (same-floor orders with $d_o = s_o$ are admissible; they contribute no elevator demand and are excluded from the $I(W)$ computation)

$r_o \in \mathbb{R}_{\geq 0}$: release time of order o

Wave-release control (operator-set):

$W_{\min}, W_{\max} \in \mathbb{Z}_{\geq 1}$: lower and upper bounds on wave cardinality

$\tau_\omega \in \mathbb{R}_{>0}$: start time of release window of wave ω

$\Delta \in \mathbb{R}_{>0}$: length of release window

Stochastic specification:

$\sigma_{M_3} \in \{0.10, 0.20\}$: lognormal noise scale on the timing primitives under model M_3

Under elevator models M_1 and M_2 all parameters above are deterministic; under M_3 , the timing primitives τ_s , τ_e , and τ_d become lognormally perturbed random variables with scale parameter σ_{M_3} . Specific numerical values for simulation experiments are reported in Section 5.

Decision variable

The model has a single decision variable, the tactical wave composition.

• $x_o \in \{0, 1\}, \forall o \in \mathcal{O}$: wave-inclusion indicator, with $x_o = 1$ if order o is released in the current wave.

The wave composition is the induced set $W = \{o \in \mathcal{O} : x_o = 1\}$. The operational layer carries no decision variables: order-to-AMR assignment, elevator boarding, and dispatch sequencing are fixed by the policy of Assumptions A1–A3, and the wave make-span $C_{\max}(W; M)$ is realized through the simulator under elevator model M . We treat the operational layer as policy-fixed rather than optimized because operational lift-scheduling optimization is already well studied (e.g., Chakravarty et al. (2025)); this paper’s contribution is the upstream tactical layer, which the literature treats as exogenous.

Assumptions

The formulation rests on five assumptions, each stated explicitly, so the scope of validity is identifiable.

1. **Single-order AMR carriage.** Each AMR transports at most one order at a time. This aligns with industry practice for parcel-scale fulfillment and rules out multi-tote AMR variants, such as those studied by Qin et al. (2024).
2. **Source-before-destination sequencing.** For each served order o , the assigned AMR visits the source floor s_o before the destination floor d_o . This is a hard physical constraint, not a modeling choice.
3. **Capacity-bounded FIFO boarding (default).** AMRs join the elevator queue in arrival order and board up to c at a time under elevator model M . The destination-clustered alternative evaluated in Section 5 follows the same capacity bound and boarding semantics, differing only in the order in which queued AMRs board the next trip.
4. **Intra-floor travel collapsed into service time.** AMR horizontal travel within a floor is summarized by the per-event service time τ_s , with no explicit congestion or routing on the floor.
5. **Elevator model is a fixed structural property of the warehouse, unknown to the operator at decision time.** The elevator model $M \in \mathcal{M}$ is a property of the physical elevator subsystem (its boarding logic and trip-time aggregation) and does not depend on the wave being released; in this sense it is exogenous to the wave-composition decision. We do not assume that the operator can identify M online. The Model-Dominance Hedge Rule of Section 4.2 is the operator's response to this epistemic uncertainty: it selects a single wave-composition decision robust to the entire family $\mathcal{M}$, without ever fitting M to data.

Structured-feature representation

Rather than working in the raw $|\mathcal{O}|$ -dimensional decision space, we summarize each candidate wave by a three-dimensional fingerprint $\Phi(W) = (C(W), I(W), T(W))$. We emphasize that $\Phi(W)$ is a deterministic function of the decision variable, not an independent decision variable: it is the representation on which our analytical tools operate, while the decision itself remains the binary vector $\{x_o\}$. The three axes are defined as follows: each captures a distinct physical mechanism of vertical-resource stress.

1. **Vertical activity diversity** $C(W) \geq 0$: It is the Shannon entropy, in natural logarithm, of the empirical floor distribution taken over the union of source and destination floors of orders in W .

With $p_f(W) = \frac{|\{o \in W: s_o = f\}| + |\{o \in W: d_o = f\}|}{2|W|}$, define $C(W) = 1 - \sum_{f \in \mathcal{F}} p_f(W) \ln p_f(W)$, with $0 \ln 0 = 0$.

High C indicates that orders span many floors, so vertical activity is diversified; low C indicates concentration on few floors. We treat C as a conceptual measure of vertical activity rather than a normalized index on $[0, 1]$.

2. **Directional imbalance** $I(W) \in [0, 1]$: It shows how one-directional the wave's cross-floor vertical traffic is, computed over the cross-floor portion of W . With $W^\uparrow = \{o \in W: d_o > s_o\}$ and $W^\downarrow = \{o \in W: d_o < s_o\}$, where same-floor orders are excluded from both sets, $I(W) = \frac{|W^\uparrow| - |W^\downarrow|}{|W^\uparrow| + |W^\downarrow|}$ when $|W^\uparrow| + |W^\downarrow| > 0$, and $I(W) = 0$ otherwise.

3. Temporal clustering $T(W) \geq 0$: It shows how tight orders arrive in time. $T(W) = \frac{\sigma_r(W)}{\mu_r(W)}$ is the

coefficient of variation of per-order release times, where $\mu_r(W) = \frac{1}{|W|} \sum_{o \in W} r_o$ and $\sigma_r(W) =$

$[\frac{1}{|W|} \sum_{o \in W} (r_o - \mu_r(W))^2]^{\frac{1}{2}}$, with r_o measured relative to the wave's release-window start. High

T means orders arrive in tight bursts; low T means they spread evenly across the window.

The three axes are orthogonal: multi-floor spread, up-versus-down asymmetry, and temporal bunching. We treat Φ as a conceptual decomposition rather than a predictive surrogate. Each axis carries a clear physical unit and operational meaning.

Elevator models

The dispatch policy is realized through a simulator under one of three elevator models:

1. M_1 (throughput aggregation): each elevator is modeled as a server with a per-AMR throughput rate that incorporates batched-trip efficiency into a single multiplier.
2. M_2 (true co-occupancy batching): at each elevator, the next c AMRs in the queue board together and ride as a unit, with realistic per-trip capacity, dwell time, and direction handling.
3. M_3 (stochastic batching): it extends M_2 with lognormal noise on per-trip duration at scale σ_{M_3} , used for robustness sensitivity.

M_1 and M_2 are both well-attested in the warehouse OR literature; the structural disagreement between throughput aggregation and explicit co-occupancy is the methodological gap our Hedge Rule resolves.

Objective function

The objective function minimizes the expected wave make-span under the chosen elevator model:

$$\min_{W \in \mathcal{O}} \mathbb{E}[C_{max}(W; M, \xi)] \quad (1)$$

where $C_{max}(W; M, \xi) := \max_{o \in W} t_o^{deliver}(W; M, \xi)$ is the wave make-span, $t_o^{deliver}(W; M, \xi)$ is the simulator-realized delivery time of order o under elevator model M on realization ξ of operational randomness, and the expectation is taken over ξ . Under M_1 and M_2 all timing primitives are deterministic and ξ degenerates, so (1) reduces to the deterministic make-span $C_{max}(W; M)$; under M_3 , ξ collects the lognormal perturbations on (τ_s, τ_e, τ_d) and the expectation is non-degenerate. We retain the expectation in (1) to keep the formulation uniform across the model family $\mathcal{M}$. We adopt make-span rather than average delivery time because wave throughput in multi-story fulfilment is gated by the slowest order: the next wave cannot release until elevator capacity is freed by the current wave's completion.

Constraints

The wave composition W is subject to two classes of constraints: tactical-layer feasibility (2)-(5), imposed at the wave-composition stage, and operational-layer feasibility (6), realized through the simulator $C_{max}(W; M)$ rather than as a formal optimization constraint.

$$W_{min} \leq |W| \leq W_{max} \quad (2)$$

$$r_o \in [\tau_\omega, \tau_\omega + \Delta], \forall o \in W \quad (3)$$

$$\sum_\omega 1[o \in W_\omega] = 1, \forall o \in \mathcal{O} \quad (4)$$

$$x_o \in \{0, 1\}, \forall o \in \mathcal{O} \quad (5)$$

$$n_{aboard}(e, t) \leq c, \forall e \in \mathcal{E}, \forall t \quad (6)$$

Constraint (2) bounds wave cardinality within the warehouse's operating envelope: W_{min} prevents under-utilized release windows, and W_{max} respects picking-station throughput and elevator-queue saturation. Constraint (3) is temporal feasibility: each order's release time r_o must lie inside the wave's release window of length Δ . Constraint (4) ensures each order is released in exactly one wave over the planning horizon. We solve (1)-(6) one wave at a time: at each release window, the decision is taken over the orders not yet released, so (4) is automatically satisfied across waves. The decision object in the rest of the paper is therefore a single wave's inclusion vector. Constraint (5) defines the binary inclusion indicator, equivalently $W = \{o \in \mathcal{O} : x_o = 1\}$. The capacity bound $n_{aboard}(e, t) \leq c$ for all elevators $e \in \mathcal{E}$ and all times t is not stated as a formal optimization constraint at the wave-composition stage; it is a simulator-enforced invariant under the dispatch policy fixed by Assumptions A1–A3. We list it as (6) only for completeness of the physical model. Likewise, the AMR fleet limit $|A|$ is realized through capacity-bounded queue handling under A3 and does not appear as an optimization constraint. The wave-composition stage's only formal constraints are therefore (2)-(5): cardinality bound, temporal feasibility, horizon partition, and binary indicator.

Equations (1)-(5), together with the operational invariants (6) and Assumptions A1-A5, define the wave release coordination problem under vertical resource constraints. Direct solution by enumeration over $\{x_o\}_o$ is intractable, because the decision space is combinatorial in $|\mathcal{O}|$ and the objective is simulator-realized. Section 4 therefore introduces a three-dimensional structured representation $\Phi(W) = (C(W), I(W), T(W))$ of wave composition and develops two analytical tools that operate on it: a Bound-and-Gap decomposition theorem and a Model-Dominance Hedge Rule. The structured representation Φ is a deterministic function of the decision $\{x_o\}_o$, not an independent decision variable; its formal definition and the partition scheme on which the tools operate are deferred to Section 4.